KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN CYBER SECURITY

CYS 425-Mobile Device and Security and Forensic Test 1

Date: 4th April, 2025 **Duration:** 1Hour 30 Minutes

Instructions: *Answer all the questions.* **Marks: 35**

1. Define Mobile device forensics and explain its significance in criminal investigations. [3]
2. List three forensic tools used for mobile data extraction. [3]
3. What is Secure Boot? [2]
4. Define chain of custody. [1]
   a. Why is the chain of custody legally significant? [2]
   b. What risks arise if the chain of custody is not properly maintained? [2]

5 4. Describe the importance of preserving evidence integrity when seizing a mobile device suspected of containing financial fraud data. [2]

6 12. Name two encryption mechanisms used in mobile devices. [2]

7 b 17. Explain the legal implications of extracting data from a mobile device without consent or warrant. [2]

8 18. Describe the forensic risks of failing to disable network connectivity such as WI-FI when seizing a mobile device. [3]

9 19. Name three types of data stored in the /data partition of an Android device. [3]

10 21. A law enforcement agency seizes an Android phone from a cybercrime suspect. The device has a cracked screen, no visible damage to storage, and is running Android OS V.8.

   a. Outline the key components of a forensic report for a mobile device investigation. Include examples of the information required in each section. [10]

CamScanner

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INFORMATION AND COMMUNICATION TECHNOLOGY DEPTMENT

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PROGRAM CYBER SECURITY

COURSE MOBILE DEVICE SECURITY AND FORENSIC

COURSE CODE CYS 425

YEAR 4th

TEST ONE (1)

...le forensics

...s is the science of extracting, processing and analysing data from ...obile devices such as cell phones, Tablets, etc. to be used as evidence in criminal proceedings or court proceedings. 1.5

Significance

- Mobile forensics is important because mobile devices contain a vast amount of data which is important and can be used for criminal investigations

The following are some key areas that contain data in mobile devices

i. Communication logs - These include Text messages, call logs, etc which are important for criminal investigations

ii. Location data - These include GPS coordinates, geotagged photos and other location data which is important in tracking the exact location of individuals relating to criminal activities

iii. Multimedia data - These include photos, Videos, audio which can be used for criminal proceedings

iv. Application data - These include data from message applications, fitness tracking data or banking data which is important in legal proceedings

Question Two (2)

i. Cellebrite Universal Forensic Extraction Device (UFED)

ii. X1 Discovery Tool

iii. FTK Imager (3)

iv. Recuva

Three (3)

ure boot - This is a security feature designed to ensure only sted software is run during the boot up processing or the starting p of a device.

If ensures no malicious third party software runs on the device, it it a security mechanism designed to prevent malicious software.

(2)

Question Four (4)

Chain of custody refers to the documentation and the tracking of evidence from the time it is seized to the time it is presented in Court for legal proceedings.

(1)

a) Significance of chain of custody

i. It ensures the integrity of evidence and it prevents changes to the evidence because it keeps track of who handled the evidence, when the evidence was accessed (e.g collection)

ii. It also helps to provide reference for any accusations of misconduct during the investigations because it provides or keeps records of on each stage of who handled the evidence.

(4)

b) Risks that arise if the chain of custody is not properly maintained?

i. The evidence will lose its integrity because it can be changed or modified by anyone without any proper authorization.

(2)

ii. Accusations of misconduct may arise as a risk and the evid
not be admissible in Court for legal proceedings.

(1)

Question Five (5)

(1) i. It is important because it ensures evidence was not tampered with hence it can be used in legal proceedings

(1) ii. It also ensures admissibility in the Courts of law because the Integrity of evidence is preserved.

Question Six (6)

i. Full-Disk Encryption

(2) ii. File-Level Encryption

Seven (7)

breaches privacy laws like the data protection act because data is being accessed without proper authorization

- It is also against the Constitution of Zambia and other laws like the Criminal Procedure Code (CPC). It raises a lot of legal implications

Question Eight (8)

Risks of failing to disable Network Connectivity

- It is at risk of being accessed remotely and changes can be made to the device

- It also prevents the device from being wiped or erased because there would not be any internet or external connectivity. The risk is that the device can be erased or wiped easily and evidence could be lost.

...ion Nine (9)

... the /data partition includes the following

- System data
ii - Application data
iii - Application databases

Question Ten (10)

Key components of Forensic Report

* TITLE
- which should include the case number of the report, the date and who carried out the report.

* EXECUTIVE SUMMARY
- This is a brief explanation on the entire case, it summarizes everything

* Case Background
- This gives a background of the case at hand and what lead to it.

*Objectives
- The main goal of the report and why it was conducted
- What is the aim

* Main Body
- It includes documentation of the device such as device information e.g make, Serial number and Operating System (OS)
- It includes physical conditions of the device eg cracks
- It includes how the entire device was seized and handled and how it was stored.
- The main body also includes the tools used for the extraction such as cellebrite UFED or Reacra

* Findings
- This part includes what was found from the tools which were used

* Analysis
- Based on the findings what have you discovered or drawn out to present as evidence.

* Conclusion
- What has been discovered from the evidence, was it used as part of the crimes committed

* Recommendations
- What do you recommend based on the Investigation.